ABS Eagle CRoll

Version 2026.1

User's Guide

AMERICAN BUREAU OF SHIPPING 1701 City Plaza Dr. Spring TX 77389

Contents

Known Issues and System Requirements for the Current Release

Known Issues

- None

System Requirements

- Operating System: Windows 11 64bit
- Hard Drive Space: 2.5 GB (application installation)
- Memory (RAM): 8 GB minimum

Prerequisite Software

- .NET Framework 4.5

1 Introduction

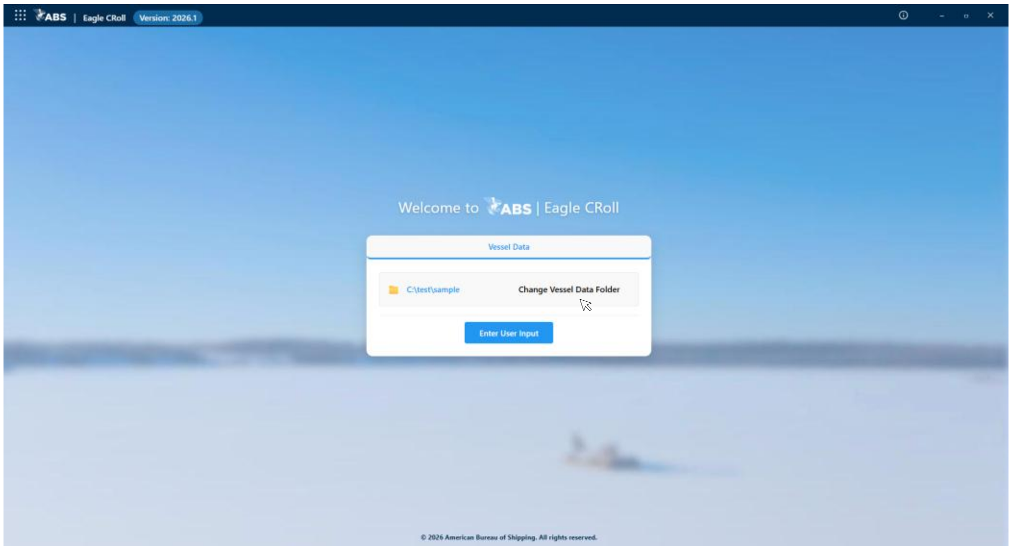
Eagle CRoll is an engineering application that enables users to visualize the roll motion magnitude for container carriers based on a precalculated vessel motion database and user-defined operating conditions. The software provides information to assist operational decision-making aiming at reducing the likelihood of parametric roll. It can also be used to support voyage planning for container carriers.

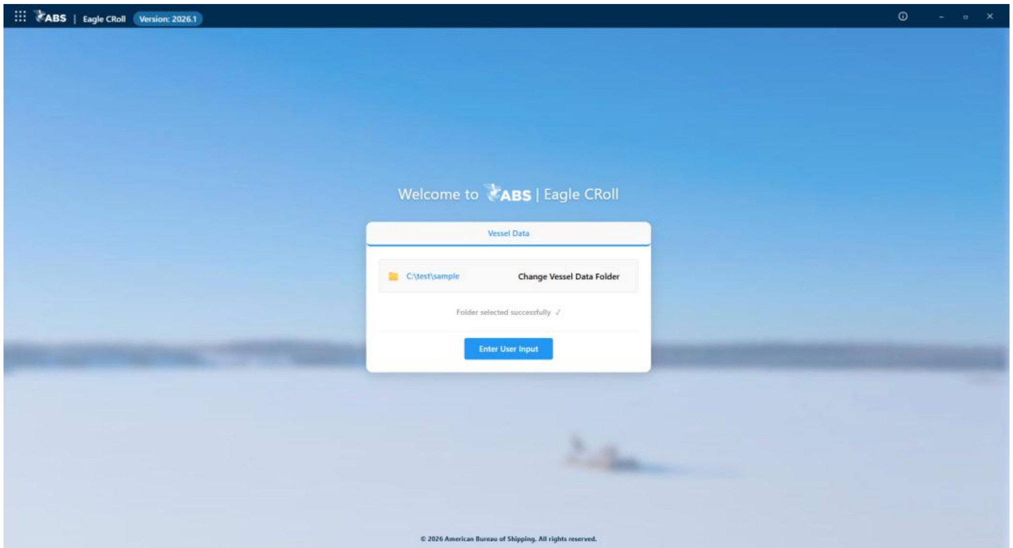
The ABS Eagle CRoll Software User Guide provides instructions for using the application, including data input, roll motion polar diagram plotting, and report creation.

2 Select Vessel Motion Database

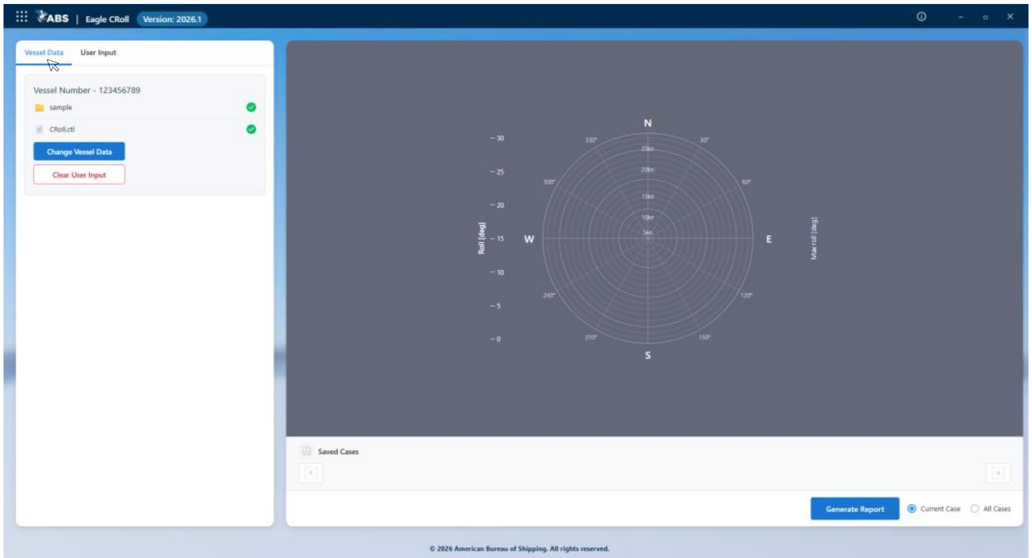
Upon launching **Eagle CRoll**, the following application home page appears.

Click **Change Vessel Data Folder** to open the File Explorer window and select the folder containing the vessel motion database. This database is precalculated using nonlinear hydrodynamic seakeeping analysis for a range of vessel conditions and sea states. Once the folder is opened, a confirmation message will appear to indicate that it has been successfully loaded.





Select the **Enter User Input** icon to open the user input window and polar diagram plotting area shown below.



The left panel displays the vessel database folder and the database control file. The control file, with the default name **CRoll.ctl**, is located in the same folder as the vessel database. Green check marks indicate that the vessel database and control file have been loaded successfully.

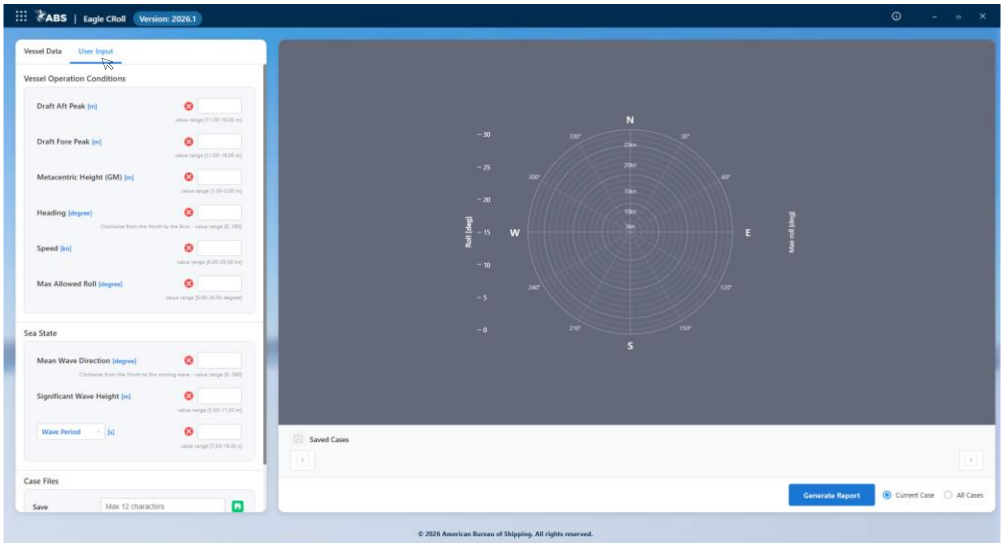
Select the **Change Vessel Data Folder** icon to return to the home page and update the vessel data folder selection.

Select the **Clear User Input** icon to delete all user input data described in Section 3.

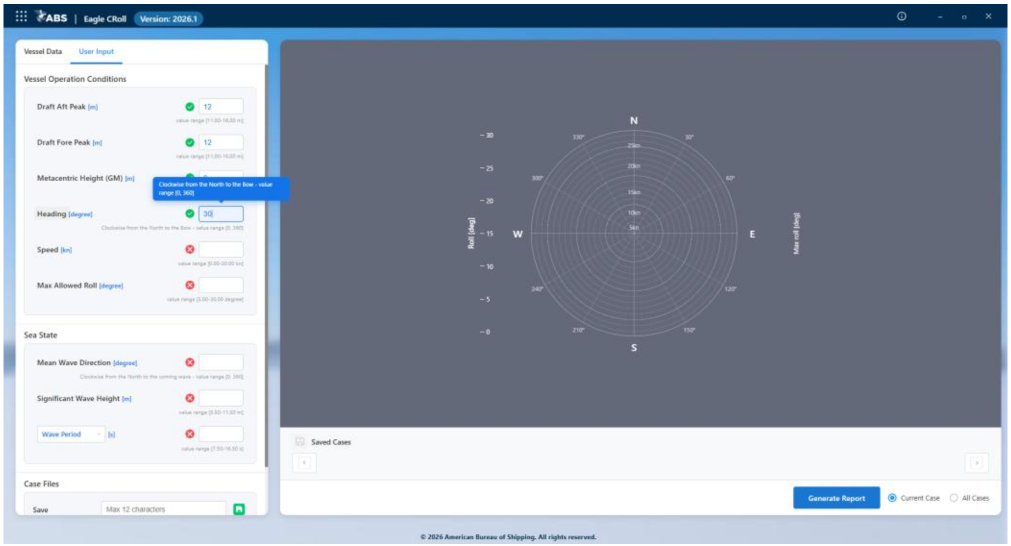
Select the **User Input** tab next to the **Vessel Data** tab to open the data input panel and enter the vessel operational data described in Section 3.

3 Enter User Input Vessel Operational Data

Select the **User Input tab** next to the **Vessel Data** tab on the left panel to open the data input panel.



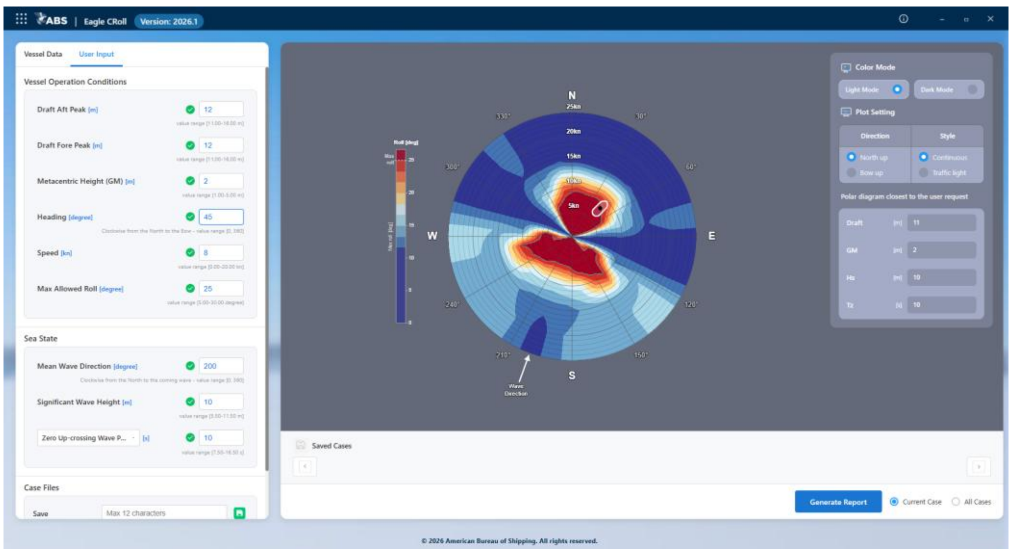
Hover over or click each input field to view the dynamic help prompt message. A red cross indicates that the entered value is outside the valid range. When the entered value falls within the valid range, a green check mark is displayed.



4 Plot Polar Diagram of Roll Motion

After all input values have been provided, the software searches the vessel database and selects the data set whose vessel draft, metacentric height, wave height, and wave period are closest to the user-entered operating conditions. The values of these parameters for the selected data set are displayed in the right-side data panel.

Using the selected data set, the software plots a polar diagram of vessel roll motion as a function of vessel heading and vessel speed. The angular axis represents vessel heading, measured clockwise from North to the vessel's bow. The radial axis represents vessel speed.



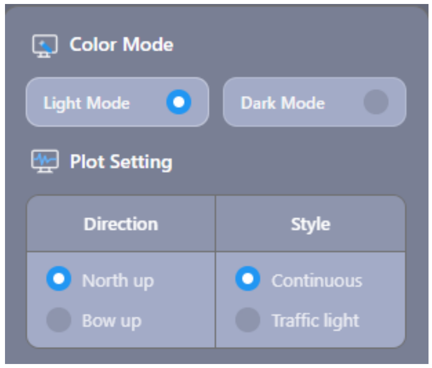
The user-defined **Max Allowed Roll** sets the boundary of the red zone in the contour plot. This boundary is highlighted by a gray contour line. Roll motion values greater than the **Max Allowed Roll** are shown in the red zone.

A wave direction indicator is displayed on the plot based on the user input for **Mean Wave Direction**, measured clockwise from North to the incoming wave direction.

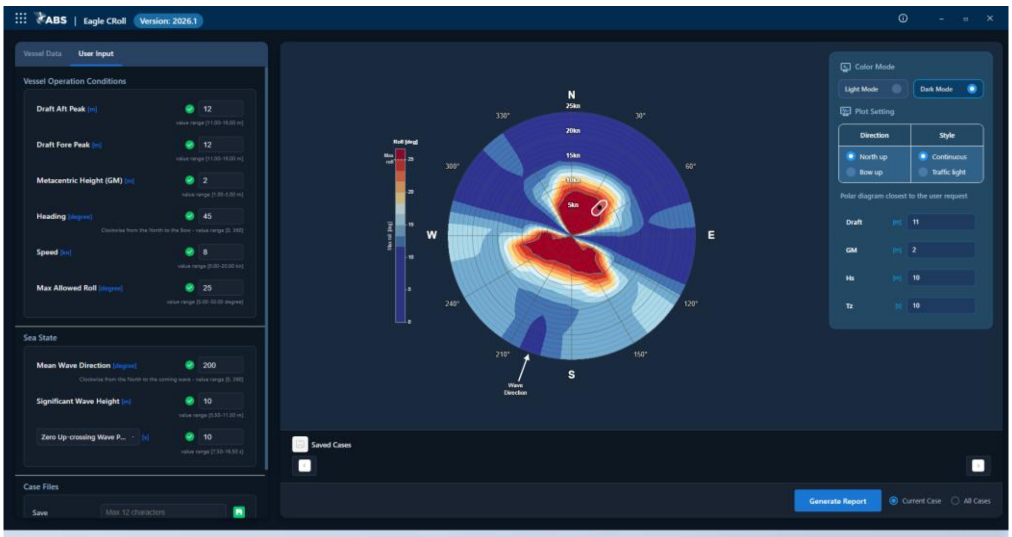
A vessel indicator, shown as a dark dot, is displayed on the plot based on the user inputs for **Heading** and **Speed**. **Heading** is measured clockwise from North to the vessel's bow. The white outline around the dark dot is a schematic representation of the vessel, indicating its bow heading.

To adjust an input value continuously, click the desired input field and place the cursor in it, then use the mouse scroll wheel or the Up and Down arrow keys. The polar diagram updates in real time as the input value changes.

The right panel provides options for adjusting **Color Mode** and **Plot Settings**.



Color Mode allows the user to toggle between **Light Mode** and **Dark Mode**.



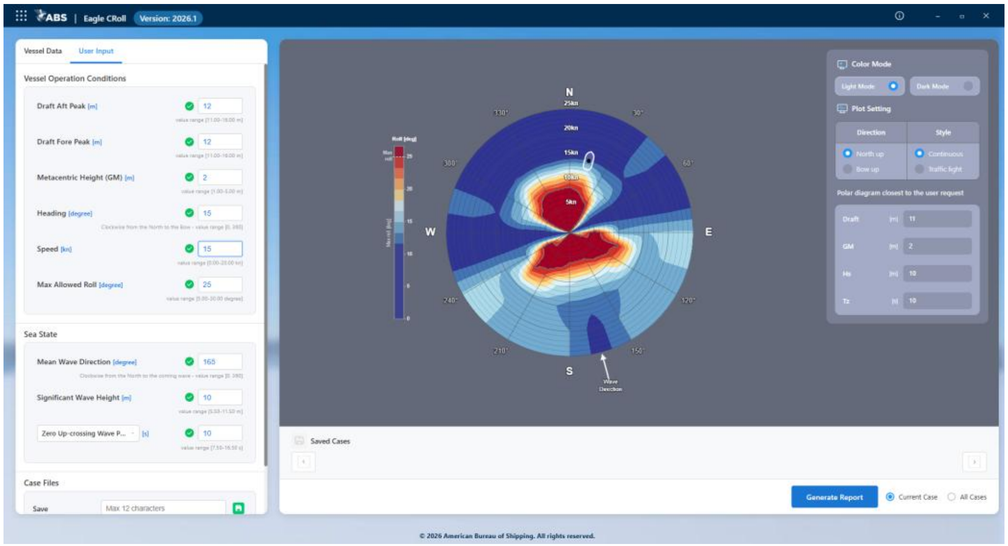
Plot Settings allow the user to adjust the display options for the polar diagram.

Direction

- **North Up** – Rotates the polar diagram so that North points upward.
- **Bow Up** – Rotates the polar diagram so that the vessel's bow points upward.

Style

- **Continuous** – Displays the polar diagram using a ten-level contour plot. The red zone indicates roll motion greater than the user-defined **Max Allowed Roll** value.
- **Traffic Light** – Displays the polar diagram using a three-level contour plot. The red zone indicates roll motion greater than the user-defined **Max Allowed Roll** value. The yellow zone indicates roll motion within 5 degrees below the **Max Allowed Roll** value.


ABS | Eagle CRoll Version: 2026.1
Vessel Data
User Input
Vessel Operation Conditions
Draft Aft Peak [m] 12
Draft Fore Peak [m] 12
Metacentric Height (GM) [m] 2
Heading [degree] 15
Speed [kn] 15
Max Allowed Roll [degree] 25
Sea State
Mean Wave Direction [degree] 165
Significant Wave Height [m] 10
Zero Up-crossing Wave P... [s] 10
Case Files
Save
Max 12 characters
Color Mode
Light Mode
Dark Mode
Plot Setting
Direction
Style
North up
Bow up
Continuous
Traffic light
Polar diagram closest to the user request
Draft [m] 11
GM [m] 2
Hs [m] 10
Tz [s] 10
N
S
W
E
Wave Direction
Saved Cases
Generate Report
Current Case
All Cases
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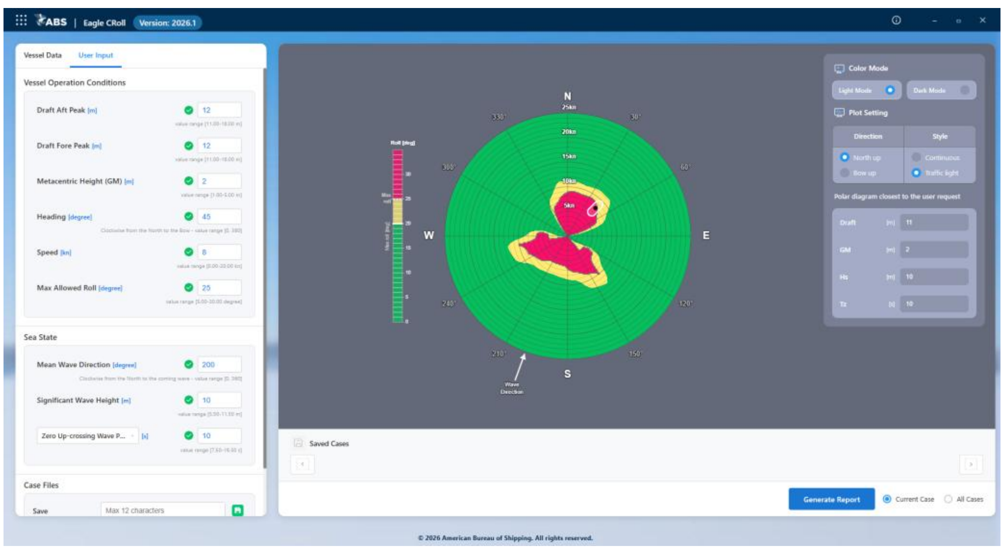
ABS | Eagle CRoll Version: 2026.1
Vessel Data
User Input
Vessel Operation Conditions
Draft Aft Peak [m] 12
Draft Fore Peak [m] 12
Metacentric Height (GM) [m] 2
Heading [degree] 45
Speed [kn] 8
Max Allowed Roll [degree] 25
Sea State
Mean Wave Direction [degree] 200
Significant Wave Height [m] 10
Zero Up-crossing Wave P... [s] 10
Case Files
Save
Max 12 characters
Color Mode
Light Mode
Dark Mode
Plot Setting
Direction
Style
North up
Bow up
Continuous
Traffic light
Polar diagram closest to the user request
Draft [m] 11
GM [m] 2
Hs [m] 10
Tz [s] 10
N
S
W
E
Wave Direction
Saved Cases
Generate Report
Current Case
All Cases
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5 Save and Delete Case Files

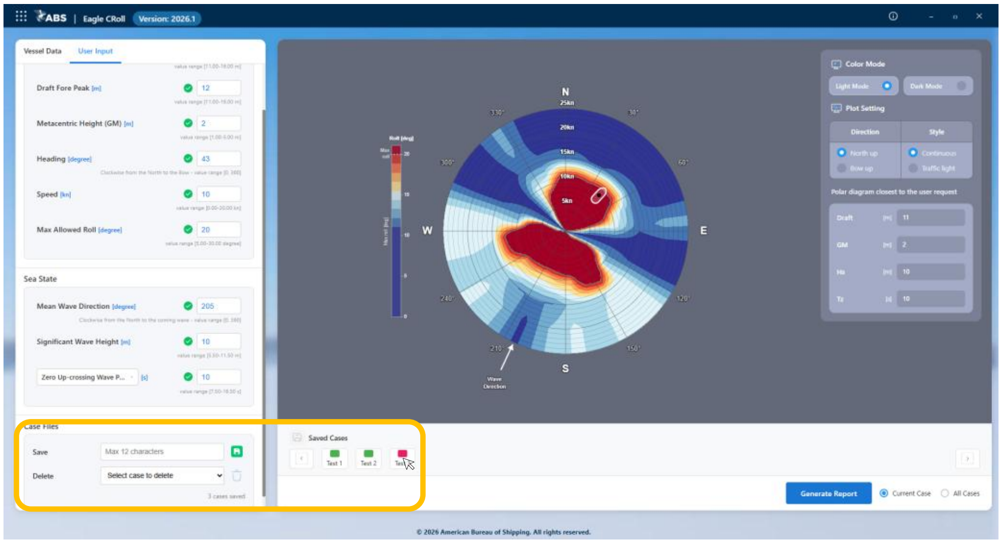
The **Case Files** section on the **User Input** panel allows the user to save the current input data and polar diagram plot as a case file. It also allows the user to delete a saved case file from the saved case dropdown list.

To save a case, enter a case file name of up to 12 characters and click **Save** icon . A new saved case icon is then added to the **Saved Cases** list below the plotting area.

The saved case icon is displayed in red when the vessel is in the red zone of the polar diagram contour plot, and in green when the vessel is outside the red zone.

Click a saved case icon to load the corresponding input data and associated plot.

When the saved case icon list exceeds the display area, a scroll bar and scroll arrows become active, allowing the user to browse through the saved case files.



6 Generate Report

Use the **Generate Report** function to create a PDF report for the currently displayed case or for all saved case files.

Select **Current Case** or **All Cases**, and then click the **Generate Report** icon to open the report preview window.

Click the **Download PDF** icon to save the report to a local hard drive.